FINITE LENGTH AT THE GENERIC POINT OF A COMPONENT AND MAXIMAL DIMENSION POINTS OF A CLOSED SUBSCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Dimension of the local ring at a generic point, finite length of a zero dimensional local ring, and independence of length. Properties, label properties-lemma-codimension-local-ring (proof line 1649); Algebra, labels algebra-proposition-dimension-zero-ring (proof line 1653) and algebra-lemma-length-independent (proof line 1655). The first gives that the local ring of the subscheme at the generic point has dimension zero, the second that such a ring has finite length over itself, and the third that the length remains finite over the larger local ring.

Source reference. Original lemma 00IX: sources/stacks-chow/tags.txt, source lines 695–695. Navigation pointer for \ref{algebra-lemma-length-independent} used inside the counted span (line 1655); the argument that consumes it is retained in the span.

Source reference. Original proposition 00KJ: sources/stacks-chow/tags.txt, source lines 753–753. Navigation pointer for \ref{algebra-proposition-dimension-zero-ring} used inside the counted span (line 1653); the argument that consumes it is retained in the span.

Source reference. Original lemma 02IZ: sources/stacks-chow/tags.txt, source lines 3318–3318. Navigation pointer for \ref{properties-lemma-codimension-local-ring}

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used inside the counted span (line 1649); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 1630); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $Z \subset X$ be a closed subscheme.*

(1) *Let $Z' \subset Z$ be an irreducible component and let $\xi \in Z'$ be its generic point. Then*
$$length_{\mathcal{O}_{X,\xi}} \mathcal{O}_{Z,\xi} < \infty$$
(2) *If $\dim_\delta(Z) \leq k$ and $\xi \in Z$ with $\delta(\xi) = k$, then ξ is a generic point of an irreducible component of Z.*

Proof. Let $Z' \subset Z$, $\xi \in Z'$ be as in (1). Then $\dim(\mathcal{O}_{Z,\xi}) = 0$ (for example by Properties, Lemma 02IZ). Hence $\mathcal{O}_{Z,\xi}$ is Noetherian local ring of dimension zero, and hence has finite length over itself (see Algebra, Proposition 00KJ). Hence, it also has finite length over $\mathcal{O}_{X,\xi}$, see Algebra, Lemma 00IX.

Assume $\xi \in Z$ and $\delta(\xi) = k$. Consider the closure $Z' = \overline{\{\xi\}}$. It is an irreducible closed subscheme with $\dim_\delta(Z') = k$ by definition. Since $\dim_\delta(Z) = k$ it must be an irreducible component of Z. Hence we see (2) holds. □